

Benchmarks for SPICE Modeling and Parameter Extraction Based on AI/ML

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Abstract—Over the past decades, the number of submitted articles that use numerical approaches for SPICE models or for characterization (extraction) of parameters of existing SPICE models has grown significantly. Many of those articles rely on synthetic data, generated either from technology computer-aided design (TCAD) or from physical SPICE model simulations; most do not model/fit measured data. Furthermore, those articles do not evaluate the physical correctness, smoothness/monotonicity, or asymptotic correctness of the approach they propose. That is sufficient for initial evaluation of proposed techniques. However, it does not prove that they are “industrial strength.” This article presents benchmarks/guidelines for the proposed artificial intelligence (AI)/machine learning (ML) SPICE modeling and characterization techniques to try to help them become practical and useful.

Index Terms—Artificial neural network (ANN) modeling, compact modeling, parameter extraction, SPICE.

I. INTRODUCTION

ACCURATE simulation of circuit behavior depends on both analysis algorithms and device models; SPICE [1] was the first to get both of these right. Historically, there have been two “camps” regarding device models. All the SPICE models, as standardized by the compact model coalition (CMC) [2], are physically based. Many models used by the power RF design community are numerical (i.e., they are based directly on measured data [3], [4]).

Physical models have some advantages: smoothness and (where needed) monotonicity; ability to model over arbitrary values of bias, temperature, and geometry, including layout dependent effects (LDEs); modeling of statistical variation, including mismatch; and modeling of noise. They also have some significant drawbacks: they can take years, even

decades, to develop (are constantly playing “catch-up” with technology [5]) and they can be difficult and/or time-consuming to extract parameters for.

Numerical models based on data tables have been proposed over a long period of time [6], [7], [8], [9]. A big advantage is that once you have data, you can directly use them, with no time needed for model development or parameter extraction. However, they have not become “mainstream” outside of power RF design.

Artificial neural networks (ANNs) have been proposed both for modeling devices [10], [11] and for parameter extraction [12], [13], [14], [15], [16]. Until now, these have not become standard practice. However, with artificial intelligence (AI) and machine learning (ML) being “buzzwords” at present, and with good open-source software for machine learning such as PyTorch [17] being available, there is renewed interest in ANNs both for device modeling and for parameter extraction.

Some of the proposed approaches are promising, especially knowledge-based neural networks (KBNNs) [18], which blend domain knowledge (which needs to be neither complete nor highly accurate) with nondomain specific numerical techniques. KBNNs enable a first-order physical model to be “corrected” to significantly improve accuracy of modeling of measured data.¹

Unfortunately, many applications of AI/ML to modeling are not “industrial strength.” They often use synthetic data, generated either from technology computer-aided design (TCAD) simulations or from a physical compact model, and for parameter extraction generally only determine a small subset of the overall parameters. That is acceptable for initial exploration and “proof-of-concept,” which has been achieved.

However, AI/ML models and approaches to parameter extraction have moved past the proof-of-concept stage. If they are to become mainstream, they should demonstrate that they work on measured data (which are affected by noise and other imperfections) which come from a wide variety of device biases, temperatures, and geometries. For parameter extraction, they also need to be shown to be able to determine complete parameter sets, not just a small subset of the overall SPICE model parameters.

A common claim of both the AI/ML models and parameter extraction procedures is that they are computationally more efficient (i.e., are “faster”) than conventional approaches.

¹The KBNN approach was recently resurrected as a “fusion” model [19].

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However, that capability is usually based on CPU times after training based on data from a small number of fixed geometries. It does not account for retraining when data from different biases, geometries, and temperatures must be fit, when data from different technologies or device flavors (threshold voltage type, pMOS or nMOS) must be fit, or when statistical variations must be fit (including tuning to process center). The proposed AI/ML technique that we are aware of, for either modeling or for parameter extraction, cannot handle those types of variation.

In other words, the proposed numerical techniques should be shown to at least have the full capabilities of the existing physical compact models, or the best-in-class extraction techniques for the parameters of such models. If they cannot match the present state-of-the-art industrial capability, they should show how they are better than previously published AI/ML approaches. Some opportunities for improving the application of AI/ML to modeling were discussed in [19]. Here we provide a more structured, concrete, and complete proposal for how to advance and make that goal happen.

This article presents benchmark tests that the proposed AI/ML models or parameter extraction procedures need to pass to demonstrate that they are “industrial strength.” Section II presents checklists that should be included in AI/ML SPICE modeling articles, to enable evaluation of the innovativeness of their contribution. The remaining sections elaborate on details of the items in those checklists.

II. EVALUATION CHECKLISTS

Tables I and II present checklists for AI/ML approaches to modeling and to parameter extraction, respectively.

An article proposing a method for AI/ML modeling or parameter extraction does not have to check all the boxes in Tables I or II.² However, it does need to demonstrate that it checks more of the boxes than previously proposed AI/ML approaches, or is significantly superior to previous approaches (that check the same boxes) in some way. This requires more than just presenting results of a proposed approach, it requires qualitative comparison to the existing AI/ML approaches.

The existing physical device models are not perfect; they may not “check the boxes” for all the items in Table I (and for parameter extraction often a sequence of programs is run, not a single extraction program that extracts all the parameters at once). However, in general, they come close. The authors of submissions on AI/ML modeling and parameter extraction should show that their proposed approach is a true step forward toward the goal of being “industrial strength.”

Many publications to date address only modeling/characterization of dc currents. For some devices, capacitance modeling/characterization is inherently linked with dc behavior modeling. For example, the forward Early effect for bipolar junction transistors (BJTs) is physically tied to base-collector capacitance, and for LD/HVMOS transistors dc modeling of the drift region critically affects proper capacitance modeling [20]. Where such interdependencies

²Development work is incremental; we recognize it will take time, and multiple steps, to close the gap between the proposed AI/ML approaches and the existing industry capability.

TABLE I
CHECKLIST FOR MODELS

capability	existing state-of-the-art	proposed AI/ML approach	best prior AI/ML approach
obeys the laws of thermodynamics	✓	?	?
accurate DC modeling for all terminal currents, on relevant log/linear scale	✓	?	?
accurate capacitance/charge modeling	✓	?	?
models DC and capacitance interaction where relevant	✓	?	?
accurate modeling of high-frequency/non-quasi-static effects where relevant	✓	?	?
works for large-signal transient simulation, including delay effects	✓	?	?
accurate noise modeling	✓	?	?
has full geometry dependence	✓	?	?
has complete temperature dependence	✓	?	?
models all necessary LDEs	✓	?	?
behaves “well” for unreasonable geometry or temperature or bias	✓	?	?
exhibits physical monotonicity over bias, geometry, and temperature	✓	?	?
is smooth (ideally C_∞ -continuous)	✓	?	?
exhibits relevant physical symmetries (currents, charges, their derivatives)	✓	?	?
exhibits asymptotic correctness over geometry, temperature, and bias	✓	?	?
includes modeling of electrothermal effects (with frequency dependence)	✓	?	?
includes, or enables, modeling of global and local statistical variation	✓	?	?
includes, or enables, model tuning to process specification corners	✓	?	?
includes, or enables, modeling of aging	✓	?	?
enables modeling of parasitics for different layouts	✓	?	?
is verified to converge reliably in at least one circuit simulator	✓	?	?

exist accurate, self-consistent modeling of both the capacitance and dc behavior should be demonstrated.

One box that the proposed AI/ML models must check is the first in Table I: they must obey the laws of thermodynamics. Specifically, when zero voltage bias is applied to every terminal of a device, the terminal currents must be zero.³ For any applied bias, a model must be passive and not generate power, both under dc and small-signal excitation. Some previous numerical approaches to modeling violate this requirement, which is why we explicitly include it.

III. BASICS

Circuit simulation models must accurately represent the large-signal (transient) behavior of devices; small-signal

³This requirement is not just for physical correctness; models that violate the requirement cause convergence problems during circuit simulation. This is one reason for the last check list item in Table I.

TABLE II
CHECKLIST FOR PARAMETER EXTRACTION

capability	existing state-of- the-art	proposed AI/ML Approach	best prior AI/ML approach
works with measured data	✓	?	?
handles noise and abnormalities in data	✓	?	?
extracts DC, capacitance, and high-frequency model parameters	✓	?	?
handles interaction between DC and capacitance behavior	✓	?	?
extracts full geometry dependence	✓	?	?
handles local variation in data when characterizing geometry dependence	✓	?	?
extracts temperature dependence	✓	?	?
adapts to biases, geometries, and temperatures of available data	✓	?	?
handles different biases, geometries, temperatures for DC, $C(V)$, RF, noise	✓	?	?
extracts parameters for self-heating	✓	?	?
characterizes all relevant LDEs	✓	?	?
enables tradeoffs in fitting currents, conductances, and capacitances over geometry, bias, and temperature	✓	?	?
extracts noise model parameters	✓	?	?
extracts local variation parameters	✓	?	?
extracts global variation parameters	✓	?	?
extracts parameters for process corners	✓	?	?
extracts parameters from limited datasets (common for 1/f noise)	✓	?	?
hand-shakes with extracted parasitics	✓	?	?

models are derived from this.⁴ Therefore, models must in general be formulated in terms of transport currents $I(V)$, nodal charges $Q(V)$, and noise current power spectral densities S_i , over bias, temperature, geometry (which includes proximity to adjacent devices, a.k.a. LDEs), and statistical variations. It is possible to use a $C(V)$, rather than $Q(V)$, formulation for two-terminal capacitances. However, for nonlinear capacitances that can lead to numerical errors during transient simulation [22].

A model should be implemented in Verilog-A, or should describe a pathway for implementation in Verilog-A.

Over the past three decades, model improvements have in part been driven by recognition of the need for smoothness, preferably C_∞ continuity but at least k th-order continuity for simulation of k th-order harmonics, and (for symmetric devices) symmetry. For MOS transistors, a range of benchmark tests have been developed over time [21], [23], [24], [25]. Any proposed numerical model for MOS transistors must pass those tests.

IV. FIT TO MEASURED DATA

Although the initial application of ANNs to device modeling was based on fitting measured data, many recent applications

⁴If not, the results of small-signal analysis will be inconsistent with the results of a small amplitude harmonic balance analysis [21], [22].

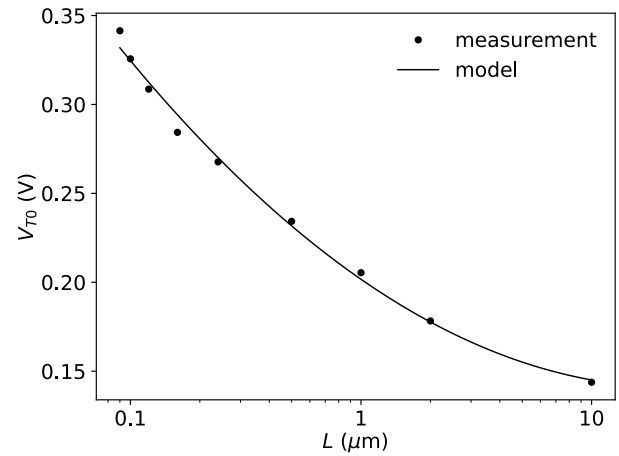


Fig. 1. Threshold voltage V_{T0} versus channel length L . The data are from measurements of a single site on a wafer, so include local statistical variation.

and analyses have been based on fitting “synthetic” data, from either TCAD simulations or compact model simulations. That is adequate for proof-of-concept and for training. However, now that the basic theoretical approach has been established, the concept needs to be shown to work on measured data and to be robust for industrial application.

Synthetic data are not compromised by imperfections. These include measurement noise (which especially shows up in the output conductance g_o of devices with low g_o , which is of primary importance for modeling for analog circuit design) and local variation.

Imperfections in data, from measurement (and other) noise and from local variation, are facts of life. Any proposed technique for modeling or parameter extraction based on AI/ML techniques should demonstrate that it is at least as insensitive to such imperfections as the existing methods are.

Fig. 1 shows measured and modeled threshold voltages as a function of channel length for transistors from a 90-nm CMOS technology. Note the “noise” in the data; this is from local variation. Just fitting the raw measured data here does not capture the underlying physical characteristics and behavior of the transistors.

Moreover, the seminal but often overlooked analysis of [26] showed that it is futile to add numerical complexity, such as “binning,” to try to improve the accuracy of modeling of data from one site on one wafer instead of fitting the underlying characteristics of the device. In simpler terms, measurements and devices have noise and statistical variability, so do not “overfit” the data [27], [28].

This is in fact an opportunity for AI/ML techniques: in practice, local variation is often “removed” from data by measuring multiple sites for one device, then taking the median of the measured data.⁵ If AI/ML can be used to infer an “underlying” model, or set of model parameter values, in the

⁵This needs to be done carefully for BJTs and for MOS transistors in weak inversion; do not just use the quantile calculation of whatever software you are using, make sure you use the method recommended in [29], and use the unbounded relative difference measure d_u defined in [30], which automatically handles small (polynomial) and large (exponential) variations.

presence of local variation, that could significantly improve modeling efficiency: it would automatically separate the signal from the noise.

V. ACCURACY, FITTING TARGETS, AND WEIGHTS

Many proposed approaches for modeling and parameter extraction based on ANNs and AI/ML target a few figures of merit, generally current or resistance (other relevant quantities are fit for RF modeling).

In practice, as noted in Section III, it is important to fit not just terminal currents but small-signal conductances, capacitances, noise, distortion, load—pull characteristics, etc. Standard device modeling allows weighted fitting/tradeoffs in accuracy over geometry, bias, temperature, and quantity (e.g., emphasize g_m a bit above V_T for long-channel devices for analog applications, and I_{eff} [31] for short-channel devices for digital applications).

Proposals for AI/ML-based techniques for parameter extraction should include details of how they handle these tradeoffs (which state-of-the-art parameter extractors handle well), and how they improve on, or at least are not inferior to, the existing state-of-the-art.

Again, this is in fact an opportunity for AI/ML-based parameter extraction: if the datasets used for training explicitly embody tradeoffs that balance the needs of digital, analog, and RF design, then the generated extractor would automatically “know” what practical tradeoffs to make when fitting a new dataset. That would be useful.

VI. ABILITY TO HANDLE (SOMEWHAT) ARBITRARY DEVICE GEOMETRIES AND TEMPERATURES

Standard modeling and parameter extraction techniques can handle input data from a wide variation in device geometries and measurement temperatures and biases.⁶ So, any practical approach to parameter extraction based on AI/ML techniques should also be able to handle such arbitrariness in input data.

For example, different modeling teams may lay out test structures with different geometry selections. Furthermore, there can be problems with data from one or more test device geometries that mean data from the “bad” device(s) must be discarded. Modeling or parameter extraction should be robust with respect to such unexpected changes in available data. In addition, models for technologies used for automotive applications must be accurate over a wider temperature range than technologies used for commercial or industrial applications.

VII. MONOTONOCITY AND SMOOTHNESS

Physically, the variation in behavior of devices with temperature, geometry, and bias is generally smooth⁷ and in many cases is monotonic. Therefore, models and model

⁶Within reason. If a model includes scaling over L and W but the data available for parameter extraction do not include sufficient combinations of L and W to make all the parameters “observable,” then this should be handled appropriately.

⁷With a small number of exceptions, see Section VIII for an example over geometry, and any device that exhibits a “snap-back” behavior in $I(V)$ will be discontinuous when driven with a voltage bias.

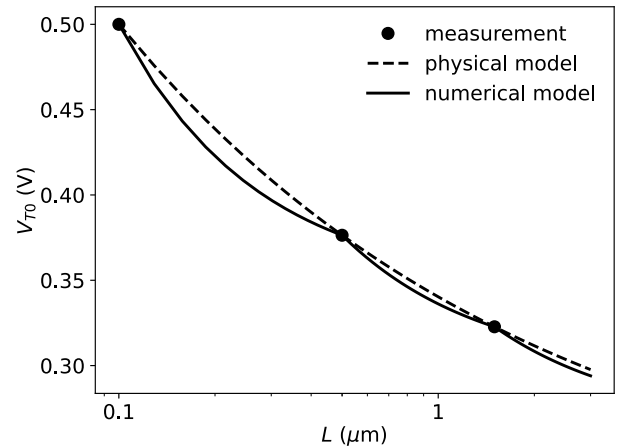


Fig. 2. Threshold voltage V_{T0} versus channel length L from a physically based and a numerical model.

parameter sets derived from AI/ML techniques should exhibit physical monotonicity and smoothness.

This is not a trivial requirement. A model is normally built, or the parameters for it are determined, based on data from a discrete grid of geometry, bias, and temperature values. What can happen, in the model, between those discrete values is not always physical.

Fig. 2 shows comparison of physical and numerical modeling of threshold voltage V_{T0} as a function of channel length L . Both are good if the goodness of fit is based solely on how well the measured data points are modeled. Between the measured data points, the numerical model is inaccurate and has unphysical behavior.

During circuit design and optimization, geometries (and biases and temperatures) other than those for which measured data are available will be evaluated. Activation functions used for AI/ML models include ReLU, GELU, and Swish. The first of these is piecewise linear, so is only C_0 continuous, and the last two are nonmonotonic. Those undesirable properties may show up in the final outputs of a model. Therefore, any proposed model based on numerical AI/ML techniques should be demonstrated to have reasonable and physical behavior for bias, geometry, and temperature values that are not in the datasets used to construct the model. This is for values both within those datasets (interpolation, demonstrated over a much finer grid of values than the input data being fit) and outside of those datasets (extrapolation).

VIII. EXCEPT WHERE PHYSICAL BEHAVIOR IS NONMONOTONIC OR DISCONTINUOUS

There are situations where device characteristics over bias, geometry, or temperature are neither monotonic nor continuous. As an example of the first, the output conductance g_{dd} of an $I_D(V_{DS})$ sweep for an “ideal” MOS transistor should monotonically decrease as V_D increases and should be positive. Because of impact ionization, g_{dd} may not monotonically decrease at high V_{DS} (low V_{GS} curve in Fig. 3). Because of self-heating, g_{dd} may become negative (moderate V_{GS} curve

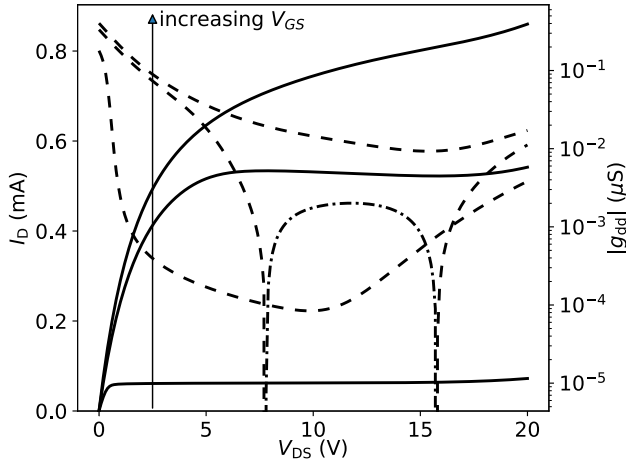


Fig. 3. Measured drain current (I_D , solid lines) and output conductance (g_{dd} , dash lines, dot-dash is where $g_{dd} < 0$) from a typical LDMOS transistor.

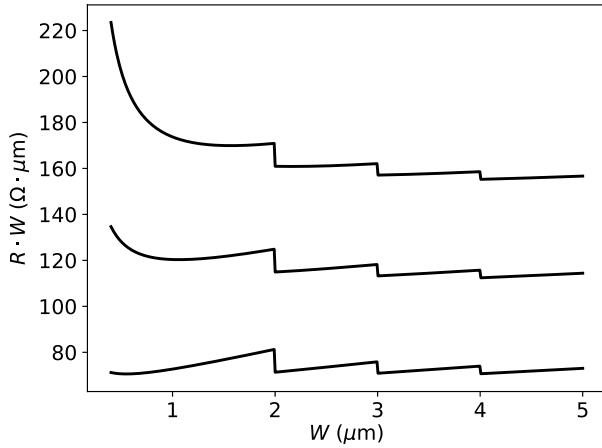


Fig. 4. Normalized resistance R versus width W for an integrated resistor. The curves, from top to bottom, are for high, typical, and low resistance process corners, respectively.

in Fig. 3; that curve is from a “static” $I_D(V_{DS})$ sweep). Any AI/ML model should be able to exhibit these behaviors.

Perhaps the simplest example of both nonmonotonicity and discontinuous behavior comes from resistors. Fig. 4 shows the resistance, normalized to width, of a resistor in a 0.18- μm BiCMOS technology.

As a first-order approximation, the resistance R of a long resistor of design length L and width W in a typical integrated circuit technology is, normalized to W , [32]

$$\begin{aligned} R \cdot W &= 2 \cdot \frac{R_C}{N_C} \cdot W + 2 \cdot R_{EW} + \frac{\rho_s \cdot L}{1 + \Delta W / W} \\ &\approx 2 \cdot \frac{R_C}{N_C} \cdot W + 2 \cdot R_{EW} + \rho_s \cdot L - \rho_s \cdot \frac{\Delta W}{W} \end{aligned} \quad (1)$$

where R_C is the resistance of a contact, N_C is the number of contacts at each end of the resistor, R_{EW} is the noncontact end region resistance per unit width, ρ_s is the sheet resistance of the body of the resistor, and ΔW is the (first-order) difference between the design and effective electrical width. For narrow widths, the last term in (1) can be dominant (depending on the value extracted for ΔW). N_C is a discrete value; incrementally

increasing W can enable an additional contact to be dropped in; from the first term in (1), this introduces a discontinuity into $R \cdot W$ versus W . For widths with the same number of contacts, the first term in (1) gives a linear $R \cdot W$ versus W variation. All these behaviors are visible in Fig. 4 and should be able to be modeled by any AI/ML approach to modeling and parameter extraction.

IX. ASYMPTOTIC CORRECTNESS

Physically, as device bias, geometry, or temperature approach extreme values, device behavior should exhibit physically appropriate behavior. This is known as the principle of asymptotic correctness [33]. This principle has often been overlooked both in the development of physically based compact models and in parameter extraction algorithms for such models.

Models are generally characterized over a finite range of biases. However, during convergence in a circuit simulator, models can be required to give reasonable values for current (and charge) for completely unreasonable values of terminal biases. Furthermore, the values for conductances, when a model is assaulted by such unreasonable biases, should be physically reasonable and should “steer” the simulator back toward a reasonable operating region.

For example, using Fourier features in a numerical model can lead to periodic/oscillatory behavior outside the bias region over which a model is characterized. This is inconsistent with the requirement of asymptotically correct behavior and does not enable a simulator to “recover” from moving to an infeasible bias point during convergence.

As another example, a KBNN approach to modeling resistors is to assume that⁸

$$\frac{1}{R} = \frac{W}{\rho_s \cdot L} \cdot [1 + f(W, L, T, V, V_G)] \quad (2)$$

where T is the temperature, V is the voltage across the body of the resistor, V_G is the voltage of the body of the resistor with respect to the layer the resistor sits in or above (the “gate,” resistors are really JFETS), and $f()$ is a function, to be “intuited” from measured data, which empirically “corrects” the simple model $R = \rho_s \cdot L / W$ (the a priori knowledge). Fitting values of R from more than a million combinations of W , L , T , V , and V_G gave an accurate model. Unfortunately, the model for $f()$ included a term in $W \cdot T$, so that as W increased outside the bounds over which the model was generated it predicted, clearly unphysically, that the current through the resistor increased as W^2 . Fig. 5 shows the difference between physically correct and incorrect asymptotic behavior.

Most modern, CMC standard, compact models exhibit asymptotic correctness. The proposed device models, from AI/ML or any other numerical approach, should demonstrate that they also satisfy the principle of asymptotic correctness.

X. LAYOUT-DEPENDENT EFFECTS

Since the end of the Dennard scaling era in the early 2000s, stress effects have been key drivers in improving

⁸This is not hypothetical, it actually happened.

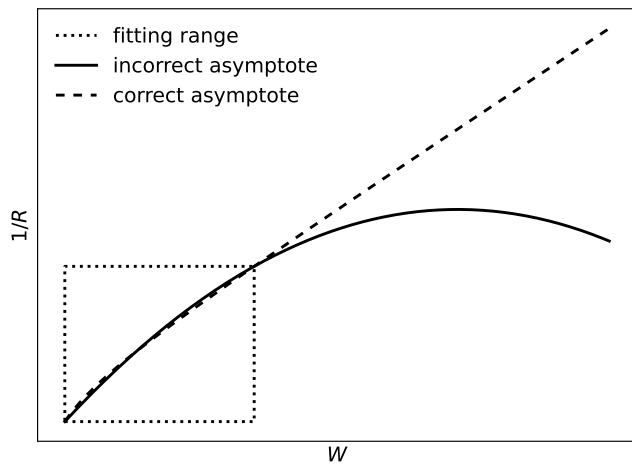


Fig. 5. Resistor conductance $1/R$ versus width W showing correct and incorrect asymptotic behavior.

MOS transistor performance. During the same time period, the proximity of a transistor to a well edge was shown to have a significant effect on transistor behavior [34]. The number of recognized LDE's has continued to increase, 14 as summarized in [35]. Scarily, although these are often modeled as being independent, [36] showed they interact. Furthermore, discerning how multiple LDEs affect device/circuit performance is difficult [37] and requires layout and measurement of a large number of test structures. The existing compact models handle LDEs via two paths.

First, some effects are handled directly, where parameters in addition to L and W are extracted from a layout and specified for a model instance [38], [39], then those are used to modify quasi-physical parameters (QFPs) of a model, such as threshold voltage V_T , mobility μ , body effect, and saturation velocity.

Second, because over the past two decades the explosion in important LDEs has outstripped the capability of compact models to keep up, the effect of LDEs is now often handled via a “black-box” interface, technology modeling interface (TMI) [40] (or the more recent open model interface (OMI), of the Si2 CMC [2]). These take (complex) layouts and compute appropriate perturbations in QFPs to pass to a model; see Fig. 6. An AI/ML-based model should be able to work within that industry standard framework.

Synthetic data, from TCAD or compact model simulations, used to date for AI/ML based modeling and parameter extraction have not included LDEs, yet LDEs are critical to modeling devices in modern technologies. The proposed AI/ML approaches for modeling should demonstrate that they encompass LDEs that are expected to be handled directly, and that they can take as input QFPs from TMI as inputs and handle them appropriately. The latter is not a simple requirement to meet: it means that a purely numerically derived model, which has no “direct knowledge” of what QFPs such as V_T and μ , should appropriately handle them as inputs that specify layout-dependent shifts in each QFP (and the implementation should demonstrate that it handles such inputs). Similarly, any proposed parameter extraction

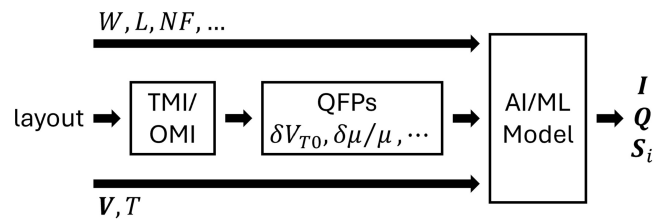


Fig. 6. Handling of LDEs.

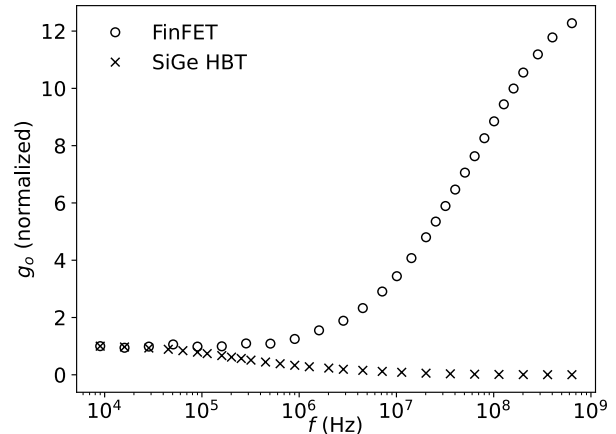


Fig. 7. Measured output conductance (normalized to the low-frequency value) as a function of frequency for a FinFET ($g_o = g_{dd}$) and a SiGe HBT ($g_o = g_{cc}$). The frequency dependence is from self-heating.

procedure based on AI/ML should demonstrate that it can characterize LDEs that are not handled by TMI.

One other aspect of LDEs: when modeling or extracting parameters for devices based on test structures, these implicitly include parasitics (capacitance, resistance, inductance where needed for RF modeling/simulation) of the exact layouts of the test structures that are measured. In practice, both modeling and parameter extraction should seamlessly work with prelayout (estimate for a typical layout) and with extracted (from an actual layout) views of a device. And both modeling and parameter extraction should account for the geometry dependence of the parasitics.

XI. SELF-HEATING

For modern technologies, many devices (BJTs, resistors, MOS transistors, HV/LDMOS transistors) are affected by self-heating, especially FinFETs and nanosheet transistors. Self-heating can give rise to seemingly “weird” behavior, such as negative g_{dd} of Fig. 3 and negative capacitances [41]. More important, the effect of self-heating is frequency-dependent, see Fig. 7. Note that in Fig. 7, both the qualitative trend (sign of change, frequency at which it occurs) and the quantitative change are device-dependent.

The AI/ML-based approaches to modeling and parameter extraction need to demonstrate that they encompass self-heating, including its frequency dependence, and in particular that they are compatible with thermal networks extracted from layout. Some AI/ML models (e.g., [42]) have included self-heating, but have not demonstrated their accuracy over frequency.

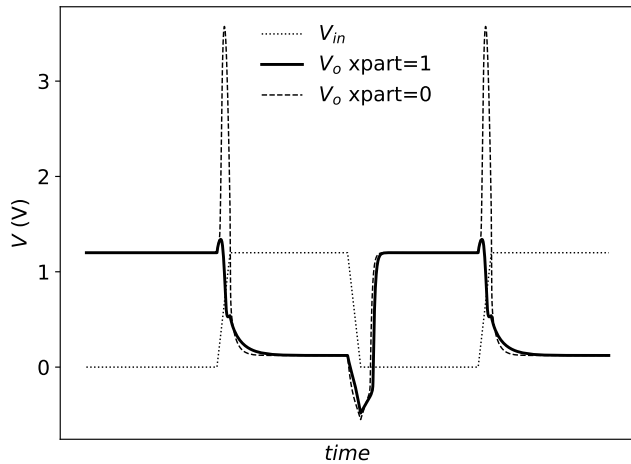


Fig. 8. Simulation of a simple inverter (with a resistive load) using BSIM4. The “spike” for $xpart = 0$ is from lack of modeling of NQS behavior. The $xpart = 1$ “correction” ameliorates the unphysical spike, but causes unphysical modeling of capacitances (see [45] for details).

XII. NOISE

Analog, RF, and digital circuits are all sensitive to noise in devices (fairly obviously for the first two; jitter in digital circuits also depends on noise). Therefore, any “industrial strength” device model should include noise.

Thermal noise should be relatively straightforward to include, it depends on effective conductance so naturally follows from the basic $I(V)$ formulation of the underlying model. However, MOS transistors with channel lengths below about 100 nm exhibit “excess” thermal noise [43].

Similarly, shot noise, where relevant, depends on $I(V)$. So, assuming implementation in Verilog-A, thermal and shot noise models should be simple to implement on top of an AI/ML-based device $I(V)$ model: that needs to be demonstrated/verified for a proposed model.

$1/f$ noise modeling is more complex, yet is critical for many circuits. $1/f$ noise depends on bias and geometry and has significant statistical variation.

The noise spectral density currents in different terminals of a device may be correlated. This is important for the input (gate, base) and output (drain, collector) terminals of MOS transistors and BJTs at high frequency.

The proposed AI/ML based models should demonstrate that they can either model all types of noise, including “excess thermal noise” and noise correlations, or that noise modeling can be “layered” on top of them. They should also demonstrate that they embody the physical scaling rules for noise defined in [44].

XIII. NON-QUASI-STATIC EFFECTS

Most compact models assume that the internal state (primarily charge distribution) of a device is in equilibrium with the instantaneously applied terminal biases (i.e., is quasi-static, QS). At high frequencies, or for switching signals with fast transition times, that assumption is incorrect, and behavior is non-quasi-static (NQS).

Sectional models can accurately represent NQS behavior, at substantial computational cost. Models based on the

relaxation-time-approximation can reasonably model NQS behavior, at moderate computational cost. For essentially no computational cost, some MOS transistor models include a charge-partitioning parameter that significantly reduces unphysical “spikes” in the transient response to input signals with fast edges, see Fig. 8. However, that completely messes up modeling of capacitances for analog simulation [45]. The relaxation-time approximation (RTA) has proven effective for first-order approximation of NQS effects. This has been added to an ANN model [46], but that does not constitute an AI/ML solution for NQS modeling.

This is another opportunity for AI/ML-based modeling: enabling accurate and efficient NQS modeling (without compromising low-frequency modeling), for both digital and analog applications at the same time, would be a significant advance over the state-of-the-art.

XIV. STATISTICAL VARIATION AND AGING

IC design involves more than verifying that a circuit “works”; it must work over specified ranges of temperature and supply voltage, it must give acceptable (profitable) yield over statistical variability, and it must maintain performance within specification, for a given aging “mission profile,”⁹ over the guaranteed lifetime of the part.

Device models must therefore encompass both statistical variation and aging. The existing SPICE models do this, either directly or indirectly, generally by implementing changes in parameters. These changes in parameters can be direct (e.g., in threshold voltage V_{T0} and gain factor k') or via mappings (e.g., from effective doping N , oxide thickness t_{ox} , and low-field mobility μ_0 into V_{T0} and k').

To be acceptable for industrial design, AI/ML-based models should also either embody statistical variations (both global and local) and aging or be able to have statistical variability and aging (often characterized via changes in QFPs) layered on top of them.

The global components of statistical process variations are often correlated between device types. For example, t_{ox} for nMOS and pMOS transistors and any capacitor that uses the same dielectric layer, or ΔL and ΔW for planar MOS transistors and poly resistors, respectively, which share the same gate polysilicon definition. These interdevice-type correlations are often not included in the existing models. If AI/ML (“big data”)-based model characterization could elicit and include those correlations, that would be a significant step forward from the present state-of-the-art.

XV. IMPLEMENTATION

A model verified outside of SPICE may be sufficient for proof-of-concept. For practical use, a model should be verified to work in one or more circuit simulators, preferably simulators that are widely used. The verification should include evaluation of convergence properties and simulation time, on a variety of typical circuits, compared with state-of-the-art, physics-based compact models.

⁹Percentage of operation at specific values of operating temperature and/or supply voltage.

This is another area where there is an opportunity for AI/ML based models. Many existing compact models include the computationally expensive log, exp, and pow operations. If a numerical model can represent measured data, with a level of accuracy comparable to that of the existing best-in-class compact models, but at significantly reduced computational cost, this would be significant.¹⁰

The present architecture of analog circuit simulators is based on how compact models are defined in Verilog-A, i.e., based on $I(V)$ and $Q(V)$ [22]. A proposed AI/ML framework for modeling devices for circuit simulation that includes an actual, or proposed, path to efficient implementation, via Verilog-A or some other mechanism, would also be significant. That framework should also specify how the implementation would interface with TMI or OMI (for LDEs, aging, etc.); see Fig. 6.

XVI. REPRODUCIBILITY, BENCHMARKING, AND ADVANCING THE STATE OF NUMERICAL MODELS

Open-source data and code play a crucial role in advancing AI/ML applications by promoting objective benchmarking and reproducibility. By providing accessible datasets and transparent codebases, open source fosters collaboration, accelerates innovation across the research community, and invites contributions from experts outside the field of device modeling. Progress in training algorithms and improved model accuracy is difficult to assess without common and extensive datasets. Reproducibility is often impossible to achieve without open-sourcing code [47].

To avoid significant or duplicate effort, one approach is to require all data and code used in a publication to be submitted along with the article. Given the importance of real-world data as described in this article, foundries and industry consortia such as the CMC can play a significant role in establishing a common database of real-world data. We advocate that such a data and code repository be established.

XVII. CONCLUSION

AI/ML-based models of semiconductor devices show significant promise, especially for characteristics such as hysteresis/memory that are hard to model physically. KBNNs in particular could be a fruitful path to follow: using a simple, physics-based core model (on which, via QFPs, geometry and temperature dependencies, statistical variation and corner models, aging, etc., can be built), with numerical techniques then used to do the “heavy lifting” to fit the difference between the simple model and experimental data. AI/ML techniques also show significant promise to determine parameter values that characterize a model for a particular technology or device.

However, the “proof-of-concept phase” for AI/ML-based modeling is over. Given modern tools such as PyTorch [17], it is not difficult to generate an AI/ML-based model or parameter extraction procedure. Many recent articles on the application of AI/ML techniques to modeling or parameter extraction essentially replicate previous work; they

demonstrate “proof-of-concept” when the concept has already been proved. What they do not do is show how they have advanced the state-of-the-art, compared with either industry best practice or the best prior work on the application of AI/ML to modeling or parameter extraction.

We provide the checklists of Section II so that future submissions on AI/ML-based models or parameter extraction procedures can show how they improve on the existing state-of-the-art AI-ML based models or parameter extraction procedures, and how they close the gap to the existing industry standard capability. In addition, hopefully, they can help guide, and provide a pathway to, future improvements.

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¹⁰Sometimes AI/ML models claim improved computational efficiency, but in general this is not for “complete” modeling and is not verified in circuit simulators on representative circuits for transient simulation, which is typically the most time-consuming type of simulation.

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